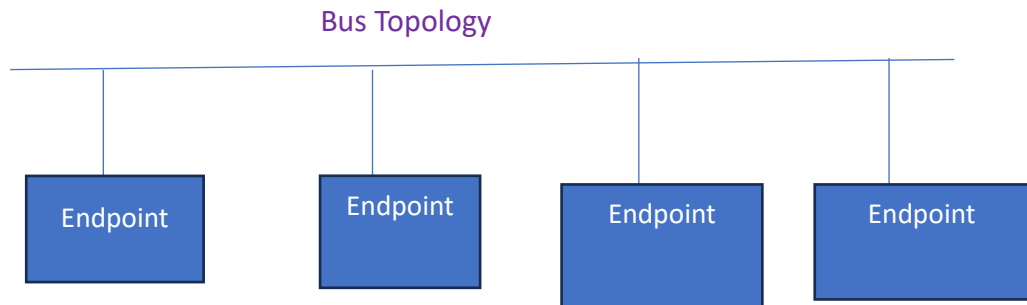


Day 2 – Network Topology

1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



Key Components

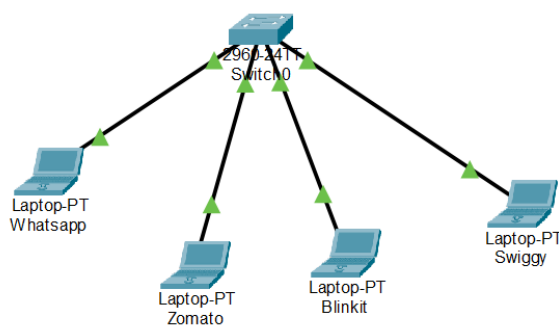
- **Main Bus Cable:** The central cable through which data travels.
- **Computers/Devices:** Connected directly to the main cable.
- **Connectors/Taps:** Connect each device to the bus cable.
- **Terminators:** Placed at both ends of the cable to prevent signal reflection.

Real-World Example

A **traditional Ethernet LAN using 10BASE2 coaxial cable** is an example where bus topology was used. It connected multiple computers through a single shared coaxial cable.

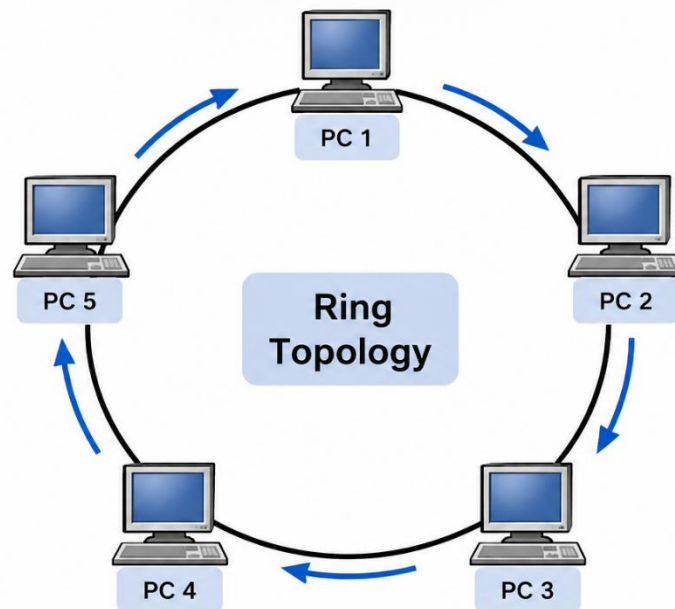
Note: Bus topology is rarely used in modern LANs because a cable failure can affect the entire network and network performance decreases as traffic increases.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



I created a **star topology** using one central switch and four laptops. The laptops are named **WhatsApp, Zomato, Blinkit, and Swiggy**, and each device is directly connected to the switch.

3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.



Data travels in one direction from one device to the next device in the ring. Each device checks the data, and if it is not for it, the data is forwarded to the next device until it reaches the destination.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages	Disadvantages
Mesh	1. Provides multiple paths between devices, so communication can continue if one connection fails. 2. Provides high reliability and fault tolerance.	1. Requires a large amount of cabling and ports, making it expensive. 2. Installation and maintenance are more complex.
Star	1. Easy to install, manage, and troubleshoot because all devices connect to a central switch/hub. 2. Failure of one device or cable usually does not affect the other devices.	1. If the central switch/hub fails, the entire network can be affected. 2. Requires more cable than simple bus or ring topologies.